

Higher priority levels (from eight to fifteen) are used for services that do not back off in response to congestion such as real-time traffic. Priority levels eight and fifteen are, respectively, for traffic that is most willing to be discarded on congestion and for traffic that is least willing to be discarded on congestion.

- Flow level (24 bits) is used to identify different data flow characteristics, which may be assigned by the source and can be used to label packets. Packet labels may be required to provide special handling of packets by IPv6 routers. One example of flow-level character is QoS or real-time services. The combination of sender IP address and flow label creates a unique path identifier that is used to route datagrams more efficiently.

Each flow is uniquely defined by the combination of the source address and a non-zero flow label. Flow label can be from 000001_h to $FFFFFF_h$ in hex. Packets having no flow label are given a zero label. All packets in the same flow must have the same flow label, same source and destination addresses, and same priority level. The initial flow label is obtained by the source using a pseudo random generator, and subsequent flow numbers are obtained sequentially.

- Payload length (16 bits) field indicates the total size of the payload in bytes of the IP datagram that excludes header fields.
- Next header (8 bits) field indicates the header that follows the IP header. The next header can either be one of the optional extension headers used by IP or the header for an upper layer protocol such as UDP or TCP. The field defines the type of extension header. For example, 0, 1, 6, 44, 51 and 80, respectively are for IP, Internet Control Message Protocol (ICMP), TCP, fragmentation header, authentication header and International Standard Organization (ISO).

Each extension header is made of two fields: extension header field and header length field. When there is no need for any other extension header, the next header will be TCP and, hence, it will contain 6. (Here, 6 indicates that there are no more extension headers. And, 6 is written in binary as 0110 in the header.) Length of the base header is fixed at 40 bytes. Extension header gives functional flexibility to IPv6 datagram. Maximum six extension headers can be used.

Extension headers are used for source routing, fragmentation, authentication, security, etc. IPv6 currently defines six extension headers: hop-by-hop option header, routing header, fragment header, authentication header, encrypted security payload header and destination options header. If one or more extension headers are used, these must be in the order as stated above. For example, if destination option, authentication and hop-by-hop option headers are to be used, the extension header fields must be: main IPv6 header, hop-by-hop option header, authentication header, destination option header and then TCP header with data.

Each extension header is associated with one 8-bit next header field. In fragment header, flags and offset fields are fixed at 16 bits. Thus, except in fragment header, in all other extension headers, the next header field is immediately followed by 8-bit extension header length. Header length indicates the length of current extension header in multiples of 8 bytes.

In the last extension header, the next header field contains the value 59. By specifying the next header field as TCP, last of the extension header can be indicated. This is because IP does not only carry TCP segment, it carries UDP pack, ICMP pack, etc, too. To standardise, the last field with value 59 is recommended.

- Hop limit (8 bits) field is used to limit the maximum number of hops that a datagram is allowed to move in the network before it

reaches its destination. If after moving the maximum number of hops, as assigned by this field, the datagram does not reach the destination, it is discarded from the network. The field is used to avoid congestion that may be caused by the datagram by repetitive and recursive movements in the network. Each router reduces the hop limit by one before releasing the datagram to the network. When the hop limit reaches zero, it is deleted by the detecting router. Hop limit of IPv6 is exactly time-to-live (TTL) in IPv4.

- IP address of the original source of the datagram is called source address. IP address of the destination of the datagram is called destination address. Each is of 128 bits. Three main groups of IP addresses in IPv6 are unicast, multicast and anycast.

Unicast address is for a particular host. A unicast packet is identified by its unique single address for a single interface network interface card (NIC) and is transmitted point-to-point.

A multicast address is for all hosts of a particular group to receive the datagram.

The anycast address is to address a number of interfaces on a single multicast address.

Unlike IPv4, IPv6 addresses do not have classes. Address space of IPv6 is sub-divided in various ways based on purpose of use. The notion for writing IPv6 address is that each double byte of the 16-byte field is separated by a colon. Bytes are written in hexadecimal symbols. Thus, one example of the address could be 1234:ABCD:56EF:0079:0000:0000:9876:456C.

The proposed notation has the provision to skip leading zeros so that 0000 can be written as 0, or 0056 can be written as 56. It also has the provision of removing one 0, leaving the colons using the technique of double colon notation.

Address of the example given above therefore may be represented as: